

PLAYWRIGHT CAPSTONE PROJECT

Project title:- Enterprise E-Commerce Framework using playwright with JS.

Website Name:- nopCommerce demo store

Website link:- <https://www.nopcommerce.com>

Project Objective:-

This capstone project focuses on building a scalable and maintainable end-to-end automation framework using Playwright and JavaScript (Node.js). The project automates critical e-commerce workflows of the nopCommerce Demo Store platform while following enterprise automation best practices such as Page Object Model (POM), reusable utilities, data-driven testing, API validations, reporting, screenshots, and parallel execution.

Project Goal:-

- Automate major user workflows of the nopCommerce demo store using Playwright and validate critical frontend functionalities.

Services / Modules used:

1)Authentication Management:-Login, logout, invalid login, forgot password, session validations

2)Product Management:- Product search, category filters, product sorting, product details validation.

3)Cart Management:-Add/remove products, quantity update, cart persistence, cart validations.

4)Checkouts and Payments Flow:-Checkout process, address forms, payment validations, order confirmation.

5)Compare and Order Management:- Product comparison, order placement, order history, invoice validation.

6)User Profile Management:-Account details, password updates, address management, account deletion.

7)Navigation and UI Validation:-Header/footer links, navigation menus, banners, responsive validations.

8)API testing and mocking :-API request validation, response verification, route mocking, network interception.

Technology Stack:-

- Playwright
- JavaScript (Node.js)
- VS code
- HTML Reports
- GitHub Version Control

Services and its Testcases Breakdown:-

1)Authentication Management	--- 15
2)Product Management	--- 20
3)Cart Management	--- 18
4)Checkouts and Payments Flow	--- 20
5)Compare and Order Management	--- 15
6)User Profile Management	--- 15
7)Navigation and UI Validation	--- 12
8)API testing and mocking	--- 10
Total Testcases	--- 125

Final Deliverables:-

- Project Testing Plan Document
- 120 Test Case Documentation
- Playwright Automation Scripts
- Test Data Files
- Bug Report / Defect Log
- Screenshots and Evidence Files
- Final Project Summary Report

Conclusion:-

This capstone project demonstrates the implementation of end-to-end automation testing on the nopCommerce Demo Store using Playwright with JavaScript.

The project covers major e-commerce functionalities such as authentication, product management, cart operations, checkout flow, compare products, user profile management, UI validations, and API testing.

A total of 125 test cases are designed across 8 different services/modules to validate the application functionality, usability, and reliability.